

Q1 (20 July 2021 Shift 2)

If the Kinetic energy of a moving body becomes four times its initial Kinetic energy, then the percentage change in its momentum will be :

- (1) 100%
- (2) 200%
- (3) 300%
- (4) 400%

Q2 (22 July 2021 Shift 1)

A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the workdone by the porter in lowering the suitcase. (take $g = 9.8 \text{ ms}^{-2}$)

- (1) -62720.0 J
- (2) -627.2 J
- (3) $+627.2 \text{ J}$
- (4) 784.0 J

Q3 (25 July 2021 Shift 2)

A force of $\mathbf{F} = (5y + 20)\hat{j} \text{ N}$ acts on a particle. The workdone by this force when the particle is moved from $y = 0 \text{ m}$ to $y = 10 \text{ m}$ is J

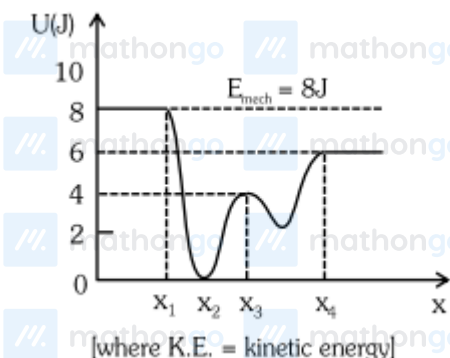
Q4 (27 July 2021 Shift 2)

Given below is the plot of a potential energy function $U(x)$ for a system, in which a particle is in one dimensional motion, while a conservative force $F(x)$ acts on it. Suppose that $E_{\text{mech}} = 8 \text{ J}$, the incorrect

Questions with Answer Keys

MathonGo

statement for this system is :



- (1) at $x > x_4$, K. E. is constant throughout the region.
- (2) at $x < x_1$, K. E. is smallest and the particle is moving at the slowest speed.
- (3) at $x = x_2$, K.E. is greatest and the particle is moving at the fastest speed.
- (4) at $x = x_3$, K. E. = 4 J.

Q5 (27 July 2021 Shift 2)

An automobile of mass 'm' accelerates starting from origin and initially at rest, while the engine supplies constant power P. The position is given as a function of time by:

- (1) $\left(\frac{9P}{8m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$
- (2) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{2}{3}}$
- (3) $\left(\frac{9m}{8P}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$
- (4) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

Hints and Solutions

MathonGo

Q1

$$K_2 = 4 K_1$$

$$\frac{1}{2}mv_2^2 = 4\frac{1}{2}mv_1^2$$

$$v_2 = 2v_1$$

$$P_1 = mv_1$$

$$P_2 = mv_2 = 2mv_1$$

$$P_1 = mv_1$$

$$\% \text{ change} = \frac{\Delta P}{P_1} \times 100 = \frac{2mv_1 - mv_1}{mv_1} \times 100 = 100\%$$

Q2

$$W_{\text{Porter}} + W_{\text{mg}} = \Delta K. E. = 0$$

$$W_{\text{Porter}} = -W_{\text{mg}} = -mgh$$

$$= -80 \times 9.8 \times .8 = -627.2 \text{ J}$$

Q3

$$F = (5y + 20)\hat{j}$$

$$W = \int F dy = \int_0^{10} (5y + 20) dy$$

$$= \left(\frac{5y^2}{2} + 20y \right)_0^{10}$$

$$= \frac{5}{2} \times 100 + 20 \times 10$$

$$= 250 + 200 = 450 \text{ J}$$

Q4

$$E_{\text{mech.}} = 8 \text{ J}$$

$$(A) \text{ at } x > x_4, \quad U = \text{constant} = 6 \text{ J}$$

$$K = E_{\text{mech}} - U = 2 \text{ J} = \text{constant}$$

$$(B) \text{ at } x < x_1, \quad U = \text{constant} = 8 \text{ J}$$

$$K = E_{\text{mech.}} - U = 8 - 8 = 0 \text{ J}$$

Hints and Solutions

MathonGo

Particle is at rest.

(C) At $x = x_2$, $U = 0 \Rightarrow E_{\text{mech.}} = K = 8 \text{ J}$

KE is greatest, and particle is moving at fastest speed.

 $x = x_3, U = 4 \text{ J}$ (D) At $U + K = 8 \text{ J}$ $K = 4 \text{ J}$

Q5

 $P = \text{const.}$

$$P = Fv = \frac{mv^2 dv}{dx}$$

$$\int_0^x \frac{P}{m} dx = \int_0^v v^2 dv$$

$$\frac{Px}{m} = \frac{v^3}{3}$$

$$\left(\frac{3Px}{m}\right)^{1/3} = v = \frac{dx}{dt}$$

$$\left(\frac{3P}{m}\right)^{1/3} \int_0^t dt = \int_0^x x^{-1/3} dx$$

$$\Rightarrow x = \left(\frac{8P}{9m}\right)^{1/2} t^{3/2}$$